Online Tech Days: AWS for Data

Implement MLOps Best Practices with Amazon SageMaker

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AI only brings value in production

By the end of **2024**, **75%** of enterprises will shift from **piloting** to **operationalizing** AI

Unclean
PoC
Experimental
Manual

Scalable
Consistent
Auditable
Reliable
Reproduceable

Source: https://www.idgconnect.com/article/3583467/gartner-accelerating-ai-deployments-paths-of-least-resistance.html

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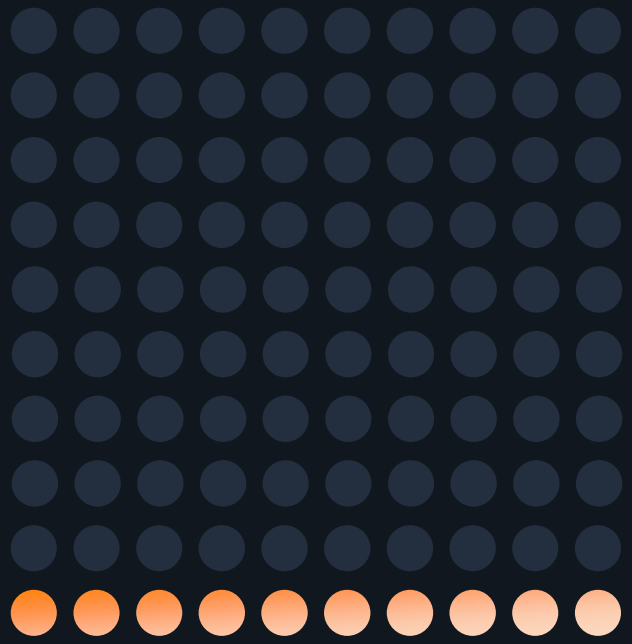
From Prototype to Production

AI Success rate in mature organizations

53%

of projects make it from prototype to production[1]

For companies still working to develop a data-driven culture



10%

of projects make it from prototype to production[1]

[1]Source: Gartner Top Strategic Technology Trends, October 2020

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ML and Path to Production

ML
Code

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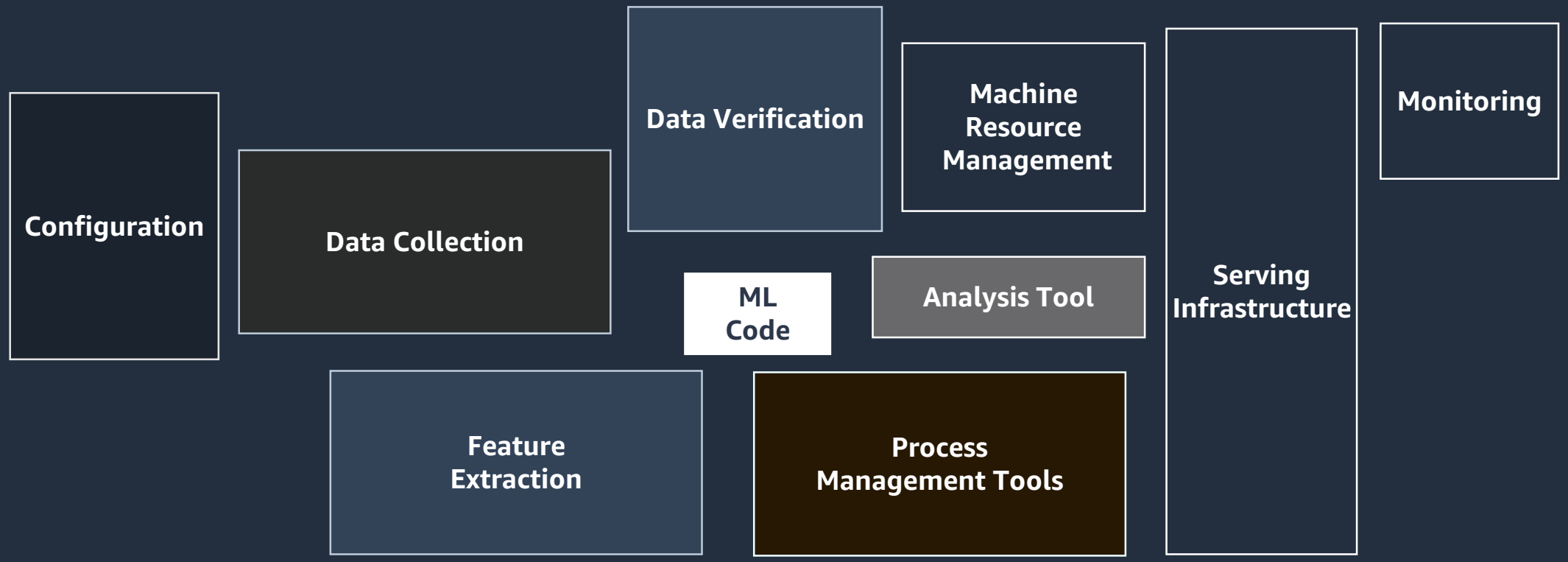
ML and Path to Production

ML CODE IS ONE SMALL PART OF THE OVERALL DEPLOYMENT PICTURE



"Only a small fraction of real-world ML systems is composed of the ML code"

source: Hidden Technical Debt in Machine Learning Systems [D. Sculley, & al.] – 2015

https://papers.nips.cc/paper/5656-hidden-technical-debt-in-machine-learning-systems.pdf

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Customer Challenges

Slow innovation pace due to operational overhead

Big gap between model building and model deployment

Lack of collaboration across different personas

Manual and unique process per business use case

No monitoring strategies in place and poor model performance

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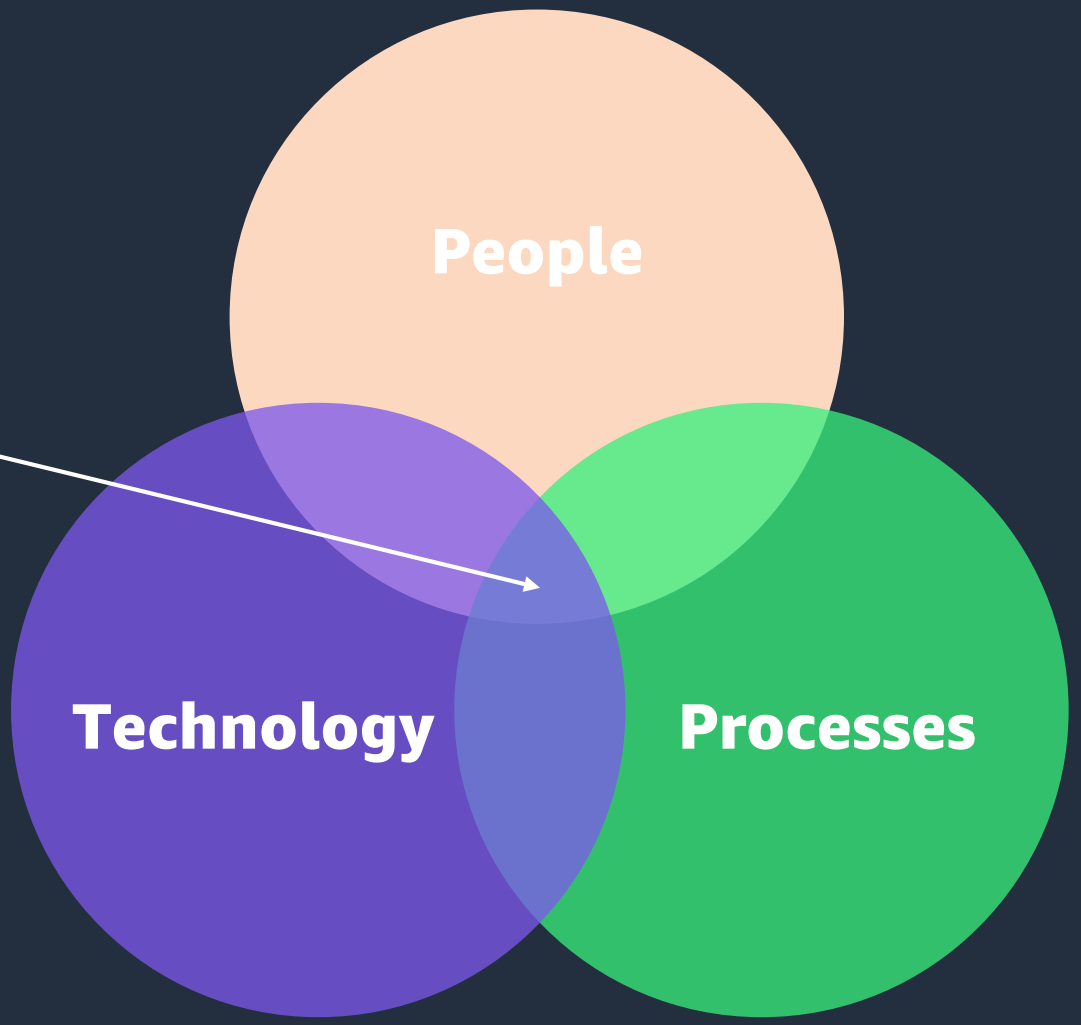
MLOps: What is it?

MLOps

"MLOps is a set of practices that aims to standardize the deployment and maintenance of machine learning models in production in order to be reliable, efficient and secure"



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MLOps Benefits: Expected Outcomes

	Business Goal	Technical Metric	Now	Expected Outcomes	Business Value
1	Be more efficient in delivery	Time to value (from research to production)	up to 12 months	< 3 months	Improve Speed-to-Value by 4x
2	Simplify route-to-live	Time to productionize existing ML use cases	3-6 months	< 2 weeks	Reduce FTE overhead in average 8x
3	Standardize infrastructure, data, & code	% Template driven development	n/a	> 85%	Focus on innovation increasing re-usability by 85%
4	Standardize onboarding of new teams and ML use cases	Time to instantiate a new MLOps infrastructure & ML projects	40 days	< 1 hours	Accelerate ML adoption across all business areas
5	Ensure high security standards	Execute the ML solutions without internet access in a private cloud	n/a	No internet	Your data is safe in your private cloud
6	**Enable Sustainability**	Reduce CO2 emissions using managed services	n/a	75%	75% greener environment with less cost
Reduce platform, people and operation costs					

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MLOps Maturity Model

KPI's

Time
To
POC

Initial → Repeteable → Reliable → Scalable

Capabilities

- **Acces to resources**
- **Standardize experimentation** environment and processes
- **Track** experiments

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MLOps Maturity Model

KPI's

Time To POC

Model deployment time

Initial → Repeteable → Reliable → Scalable

Capabilities

- **Acces to resources**
- **Standardize experimentation** environment and processes
- **Track** experiments

- **Automate** governed access to resources
- **Automate** model build, and deployment tasks
- **Standardize** source code **respositories**
- **Centralize** management of **models**

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MLOps Maturity Model

KPI's

Initial	Repeteable	Reliable	Scalable
Time To POC	Model deployment time	ML defect rate	

Capabilities

Initial

- **Acces to resources**
- **Standardize experimentation** environment and processes
- **Track** experiments

Repeteable

- **Automate** governed access to resources
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Reliable

- Integrate **full CI/CD practices** into ML pipelines for traceability, improved quality and lineage tracking

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MLOps Maturity Model

KPI's

Initial	Repeteable	Reliable	Scalable
Time To POC	Model deployment time	ML defect rate	MLDC cycle time

Capabilities

Initial

- **Acces to resources**
- **Standardize experimentation** environment and processes
- **Track** experiments

Repeteable

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Reliable

- Integrate **full CI/CD practices** into ML pipelines for traceability, improved quality and lineage tracking

Scalable

- MLDC is reproducible through **standardize templates** shared across teams

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MLOps People & Processes

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Data Science Team Structure Patterns

DATA SCIENCE TEAM STRUCTURE IMPACTS THE TECHNOLOGY STACK

Data Science Structure	Definition
Functional	Data science teams are dispersed across the organisation with a small central team
Centralised	Centralized data science team service multiple business units
Decentralised	Resources are dispersed across the organization in different silos with no view of ML activities outside their respective unit. Multiple teams exist in the organisation but are not joined together strategically

Hybrid
Functional and Centralized can co-exist

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MLOps Key Personas and Roles

Advance Analytics Team
Data Lake

Data Engineer

Prepare & Ingest data building ETL pipelines for the ML use cases

Data Owners

Manage data sharing and provide access to other teams

Data Science Team
Experimentation & MLOps

Data Scientist

Create and Maintain the best ML models to solve business problems

ML Engineer

Collaborate with DS to productionize their code and models following development best practices

Platform Team
Secure Cloud/Data/ML Platform

MLOps Engineer/ ML Admin

Standardize CI/CD, user/service role & container creation, model consumption, testing and deployment methodology based on business requirements

Security

Assess data, user, and service access creating policies and guardrails

Architects/ SysOps Engineer

Standardize account infrastructure e.g. VPC and endpoints, subnets, security groups, connectivity, user roles implementation

Risk & Compliance
Approve & Review Models

Auditors/Risk & Compliance

Review models, data sources, code artifacts, perform hypothesis tests on the models

Business
Viz Dashboards, ML Adoption, & ROI

Business Stakeholder Product Owners

Define business problem, ML use case business KPIs, interact with other BUs, and make business decisions

Business Stakeholder ML Consumers

Consumers of ML results from other BUs, driving business decision making

aws

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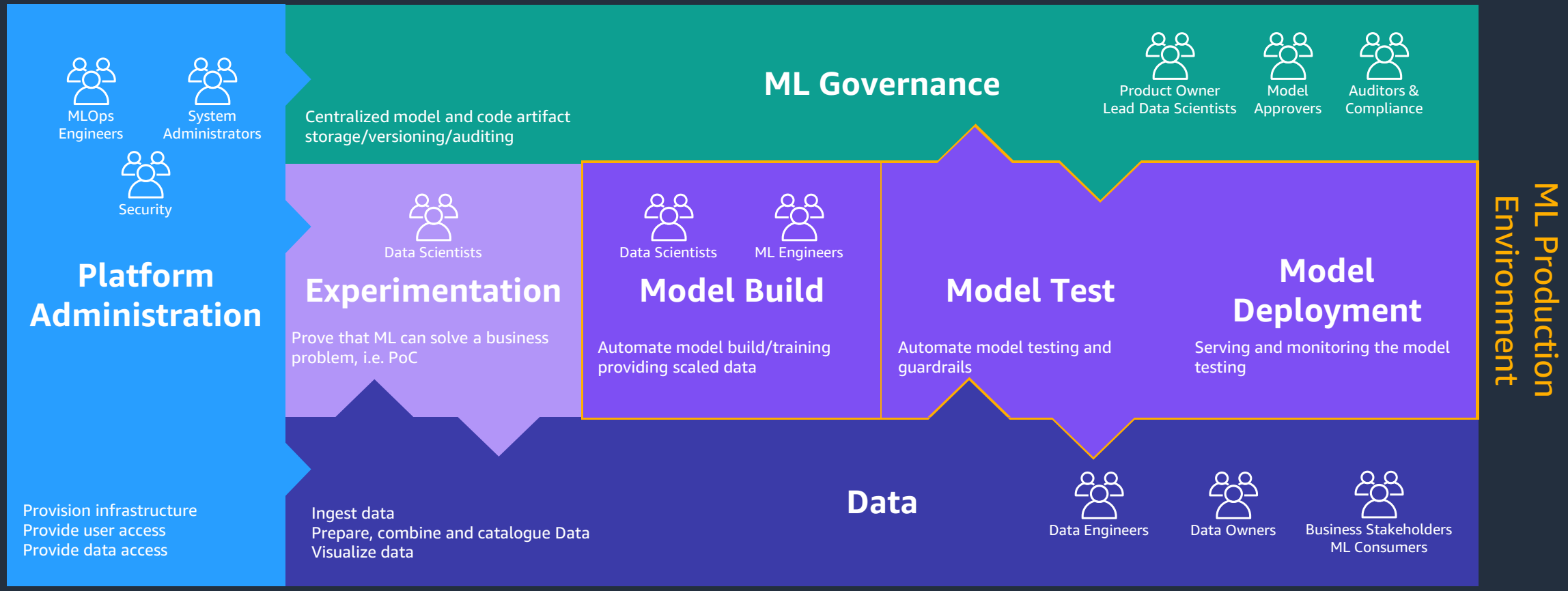
MLOps Foundation People & Processes

SEPARATION OF CONCERNS IS KEY FOR SUCCESS



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MLOps Foundation Roadmap
Technology

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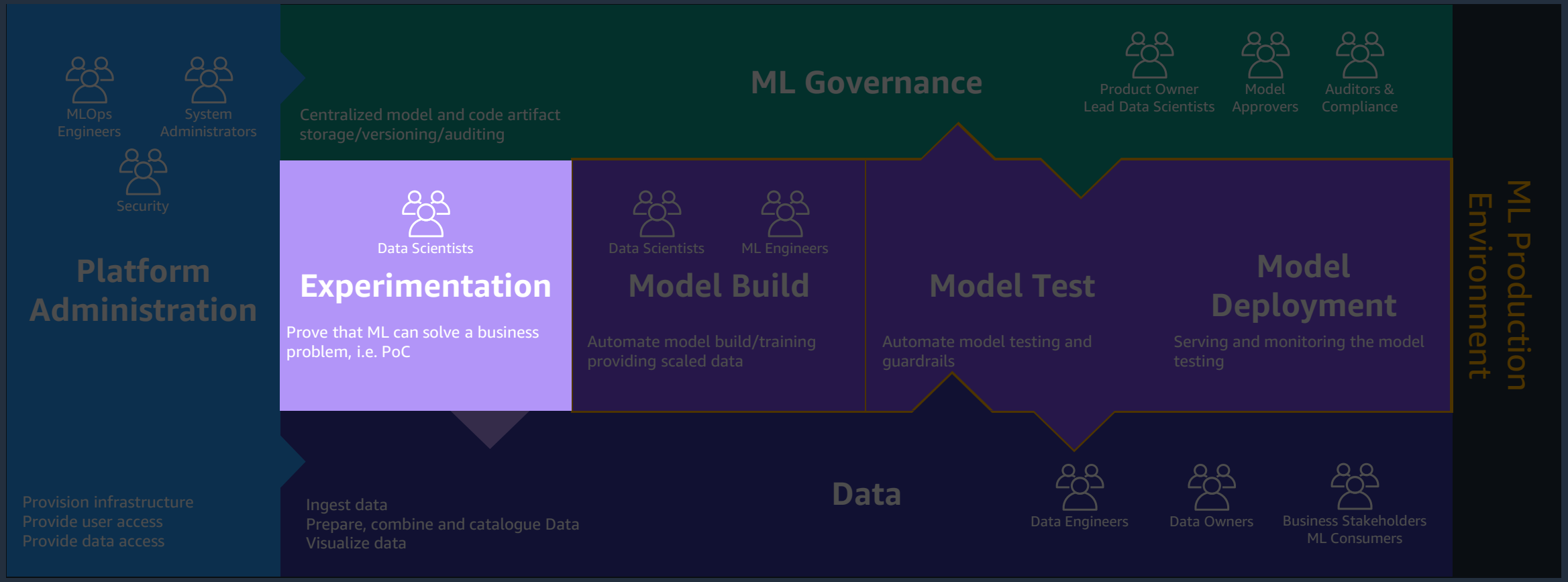
MLOps Foundation People & Processes

INITIAL PHASE



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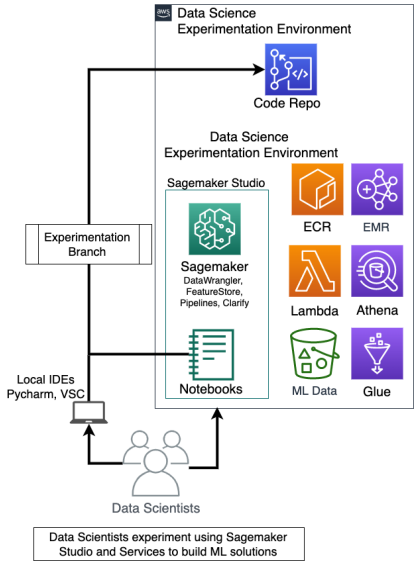
MLOPs Initial Phase

ML EXPERIMENTATION ON AWS USING AMAZON SAGEMAKER STUDIO NOTEBOOKS



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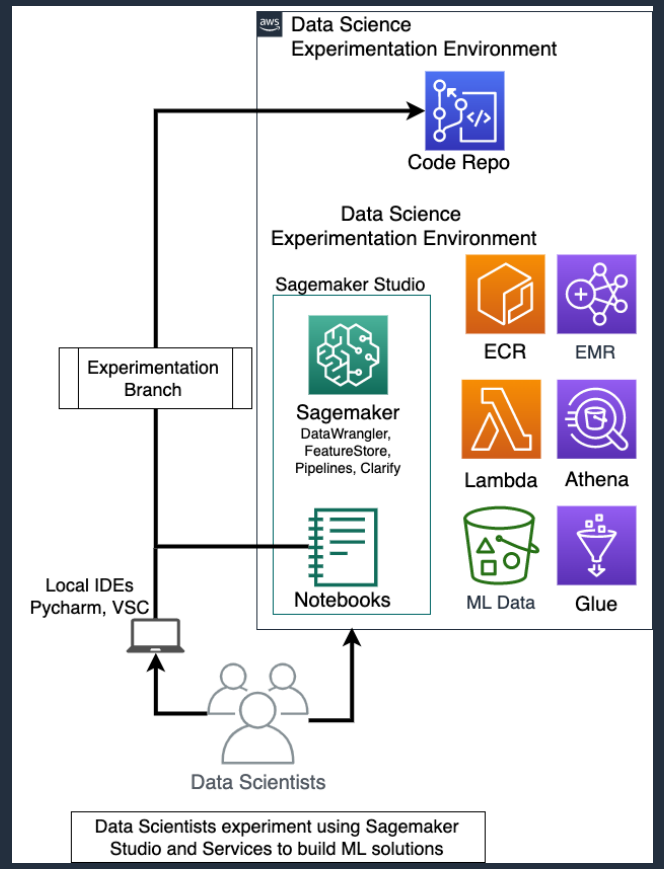
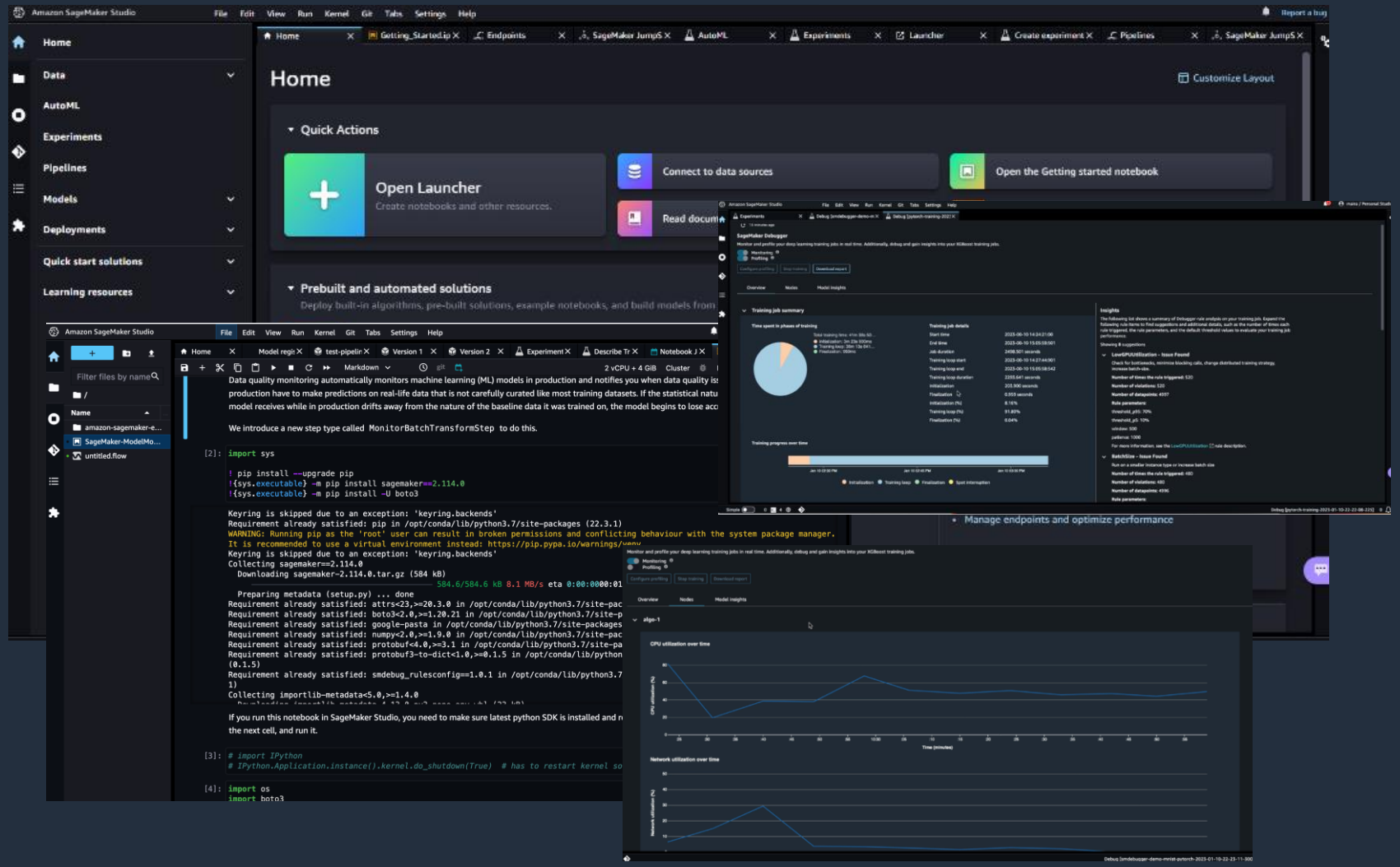
Amazon SageMaker Studio

ML EXPERIMENTATION ON AWS USING AMAZON SAGEMAKER STUDIO NOTEBOOKS





https://github.com/aws-samples/amazon-sagemaker-studio-secure-data-science-workshop

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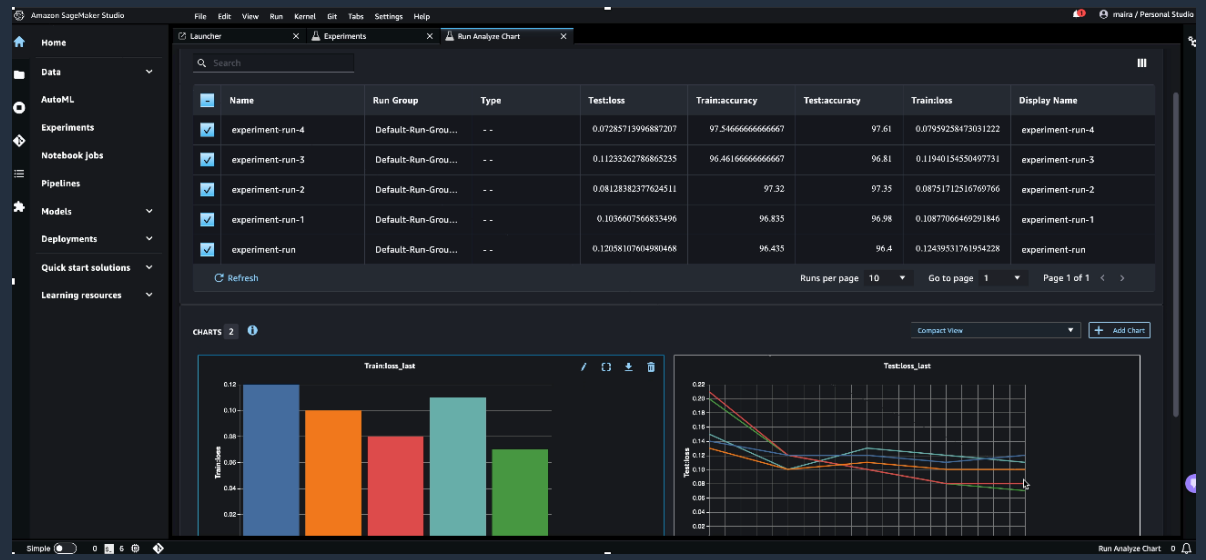
Amazon SageMaker Experiments

ORGANIZE, TRACK, AND COMPARE MACHINE LEARNING EXPERIMENTS

```python
from sagemaker.experiments.run import Run
from sagemaker.session import Session
# create an experiment, start a new run and log a parameter
experiment_name = "experiment_name"
run_name = "run_name"
with Run(
    experiment_name=experiment_name,
    run_name=run_name,
    sagemaker_session=Session()
) as run:
    run.log_parameters({
        "normalization_mean": 0.1307,
        "normalization_std": 0.3081,
     })
```

```python
from sagemaker.experiments.run import load_run
from sagemaker.session import Session
# load the run from context
with load_run(
    sagemaker_session=Session()
) as run:
    run.log_metric("precision", 0.9, step=step)
    run.log_confusion_matrix(
        target,
        pred,
        "Confusion-Matrix-Test-Data"
    )
```



Tracking at scale
Track parameters and metrics across experiments and users

Custom organization
Organize experiments by teams, goals, and hypotheses

Visualization
Easily visualize experiments and compare

Metrics and logging
Log custom metrics using the Python SDK and APIs

Fast iteration
Quickly go back and forth, and maintain high-quality

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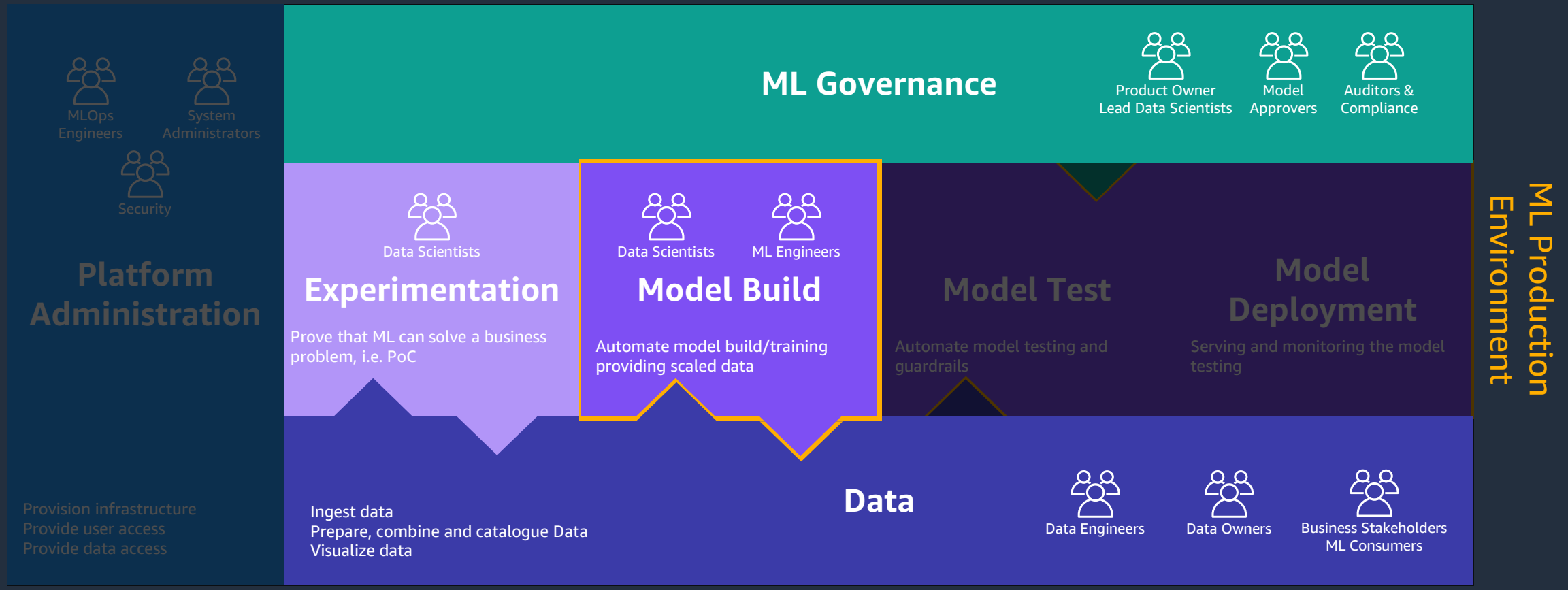
MLOps Foundation People & Processes

REPEATABLE PHASE



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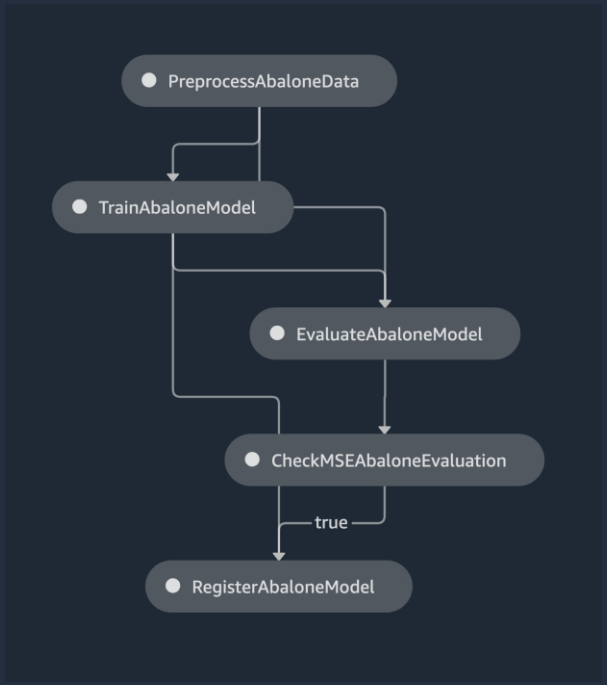
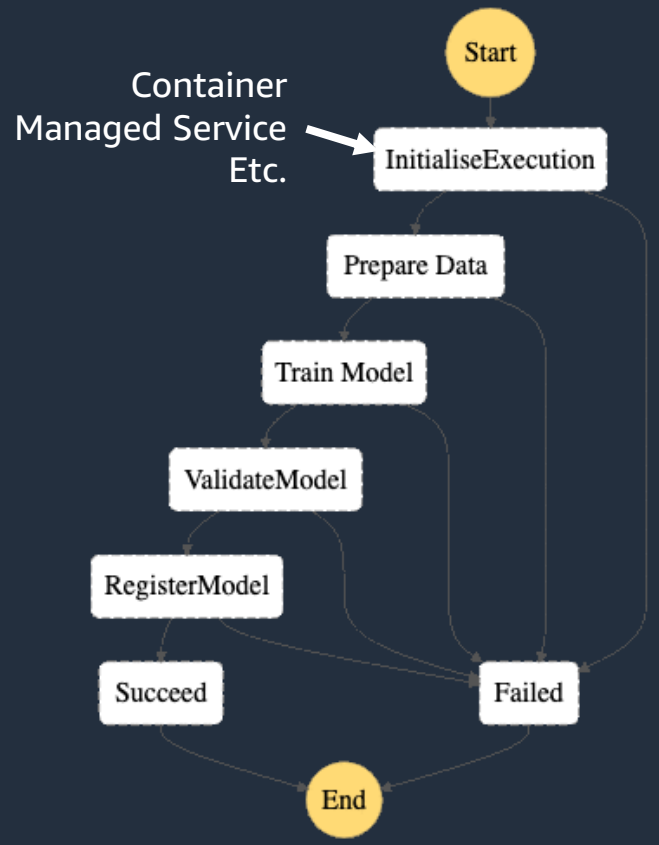
ML Solution Lifecycle Automation

CREATING AUTOMATED WORKFLOWS ON AWS – ML PIPELINES



Example model training state machine
SageMaker Pipelines



Example model training state machine and common steps

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Amazon SageMaker Processing

PREPROCESS DATA & OFFLOAD DATA SCIENTISTS ON MANAGING CONTAINERS

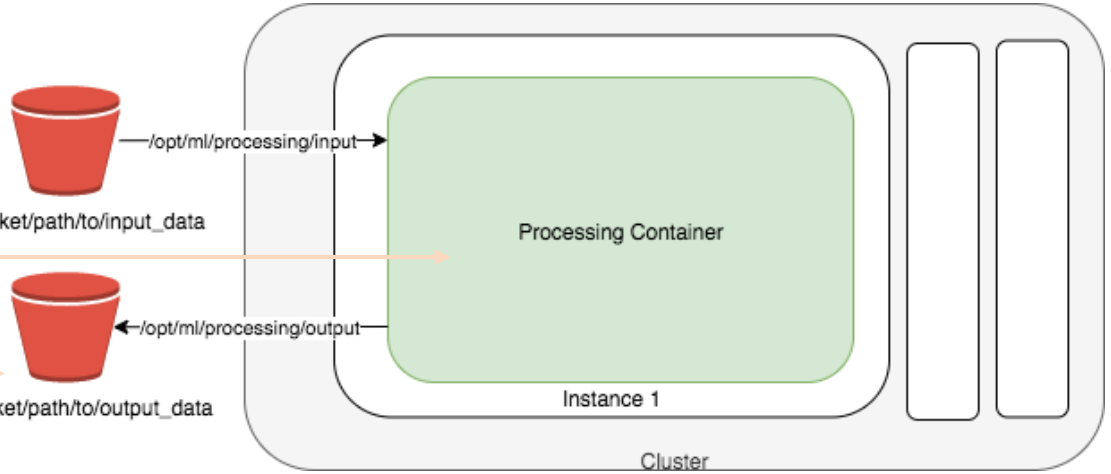
```
from sagemaker.sklearn.processing import SKLearnProcessor
from sagemaker.processing import ProcessingInput, ProcessingOutput

sklearn_processor = SKLearnProcessor(framework_version='0.20.0',
                                     role=role,
                                     instance_type='ml.m5.xlarge',
                                     instance_count=1)

sklearn_processor.run(code='preprocessing.py',
                      inputs=[ProcessingInput(
                        source='s3://path/to/my/input-data.csv',
                        destination='/opt/ml/processing/input')],
                      outputs=[ProcessingOutput(source='/opt/ml/processing/output/train'),
                               ProcessingOutput(source='/opt/ml/processing/output/validation'),
                               ProcessingOutput(source='/opt/ml/processing/output/test')]
                     )
```

Container

Code

Inputs

Outputs



Supported pre-built processors: scikit learn, PySpark, HUGGING FACE, mxnet, TensorFlow, XGBoost + Custom Containers

Framework Processors

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Amazon SageMaker Pipelines

AUTOMATED WORKFLOWS TO PRE-PROCESS, TRAIN, EVALUATION, & REGISTER MODELS

```
from sagemaker.sklearn.processing import SKLearnProcessor

sklearn_processor = SKLearnProcessor(framework_version='0.20.0',
                                     role=<role>,

from sagemaker.workflow.pipeline import Pipeline

  pipeline_name = f"AbalonePipeline"
  pipeline = Pipeline(
      name=pipeline_name,
      parameters=[
          processing_instance_type,
          processing_instance_count,
          training_instance_type,
          model_approval_status,
          input_data,
          batch_data,
      ],
      steps=[step_process, step_train, step_eval, step_cond],
  )
        ProcessingOutput(output_name="test", source="/opt/ml/processing/test")
    ],
    code="abalone/preprocessing.py"
)
```

Step Types

- Processing
- Training
- Tuning
- CreateModel
- RegisterModel
- Transform
- Condition
- Callback
- Lambda Function
- ClarifyCheck
- QualityCheck
- Amazon EMR
- Fail

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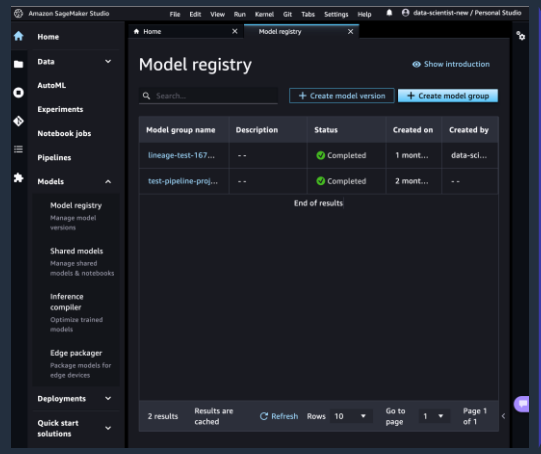
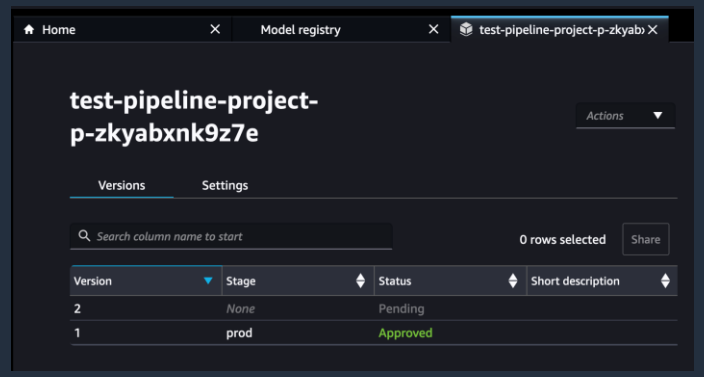
Amazon SageMaker Model Registry

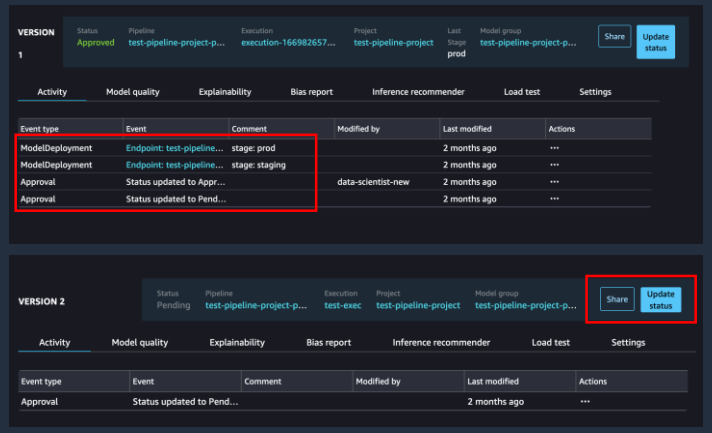
STORE, VERSION, AND TRIGGER MODEL PROMOTION



Create model groups in your model registry



Benchmark and observe the model versions in your model group and promote versions of the model by changing their status



Track the activities, metrics, and settings per model version

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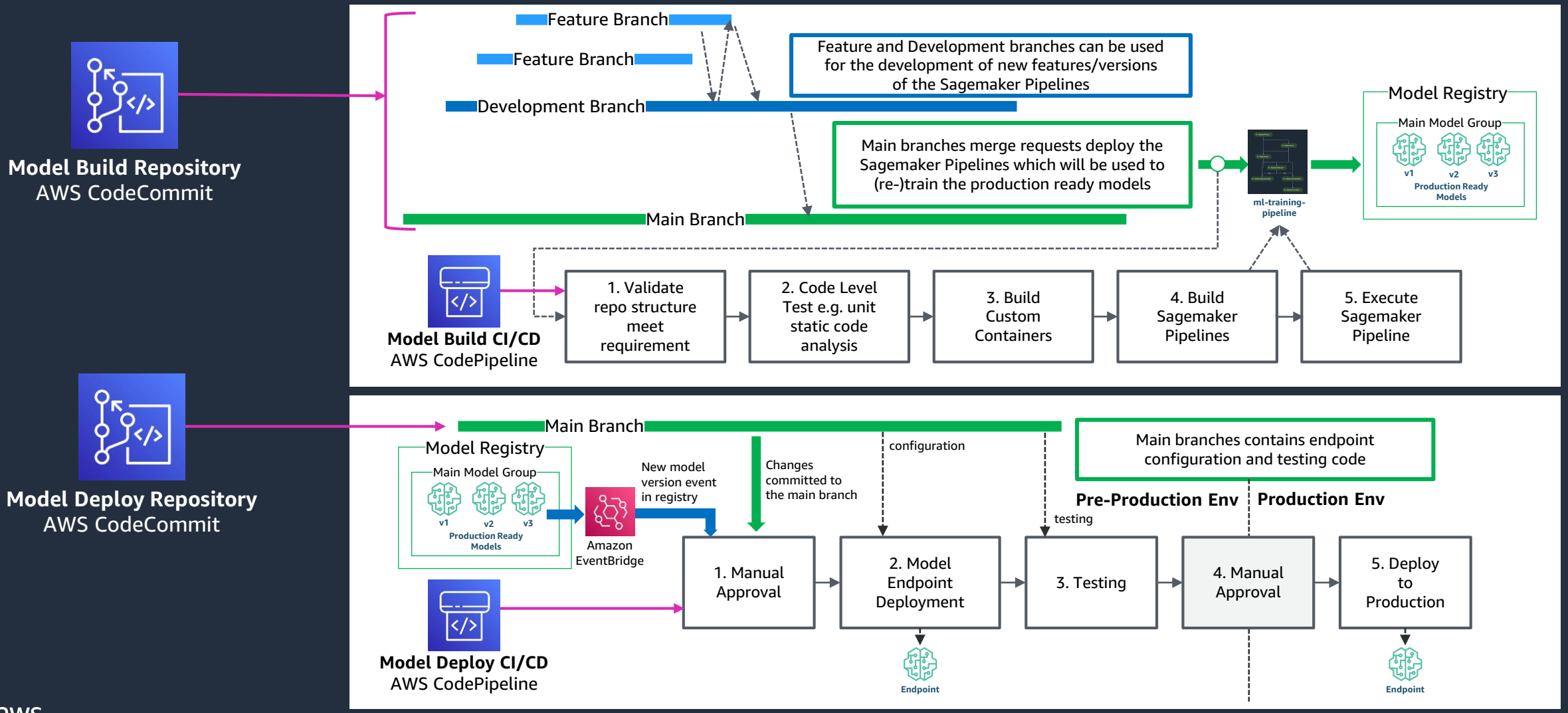
Amazon SageMaker Projects

EXAMPLE REPOSITORIES & CI/CD PIPELINES PER SAGEMAKER PROJECT



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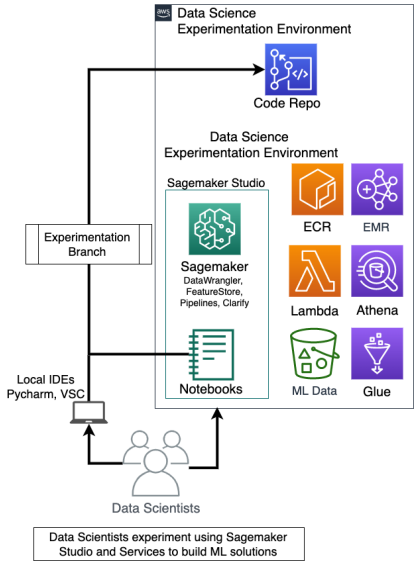
MLOPs Initial Phase

ML EXPERIMENTATION ON AWS USING AMAZON SAGEMAKER STUDIO NOTEBOOKS



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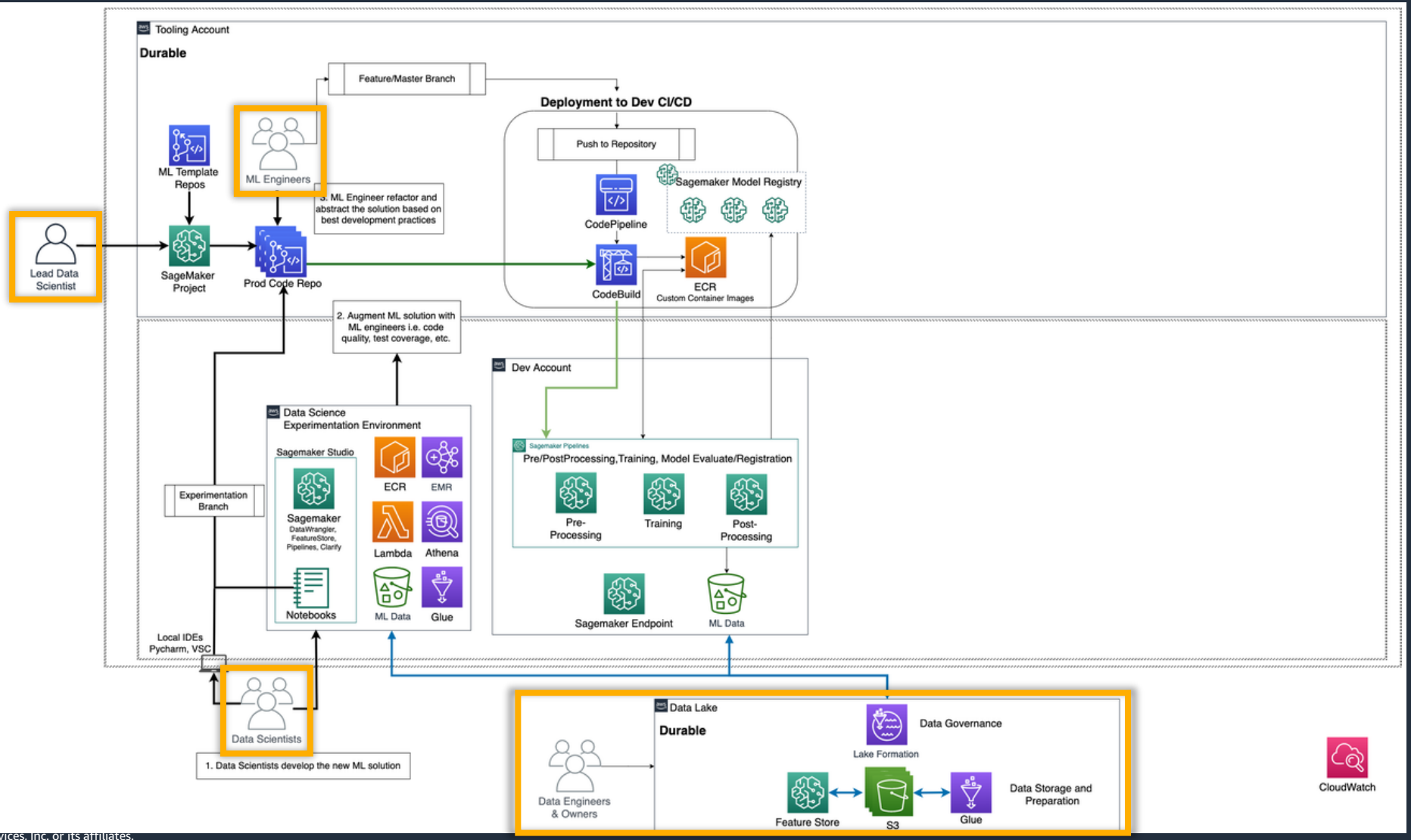
MLOPs Repeatable Phase

FROM RESEARCH NOTEBOOKS TO ML PIPELINES & AUTOMATION



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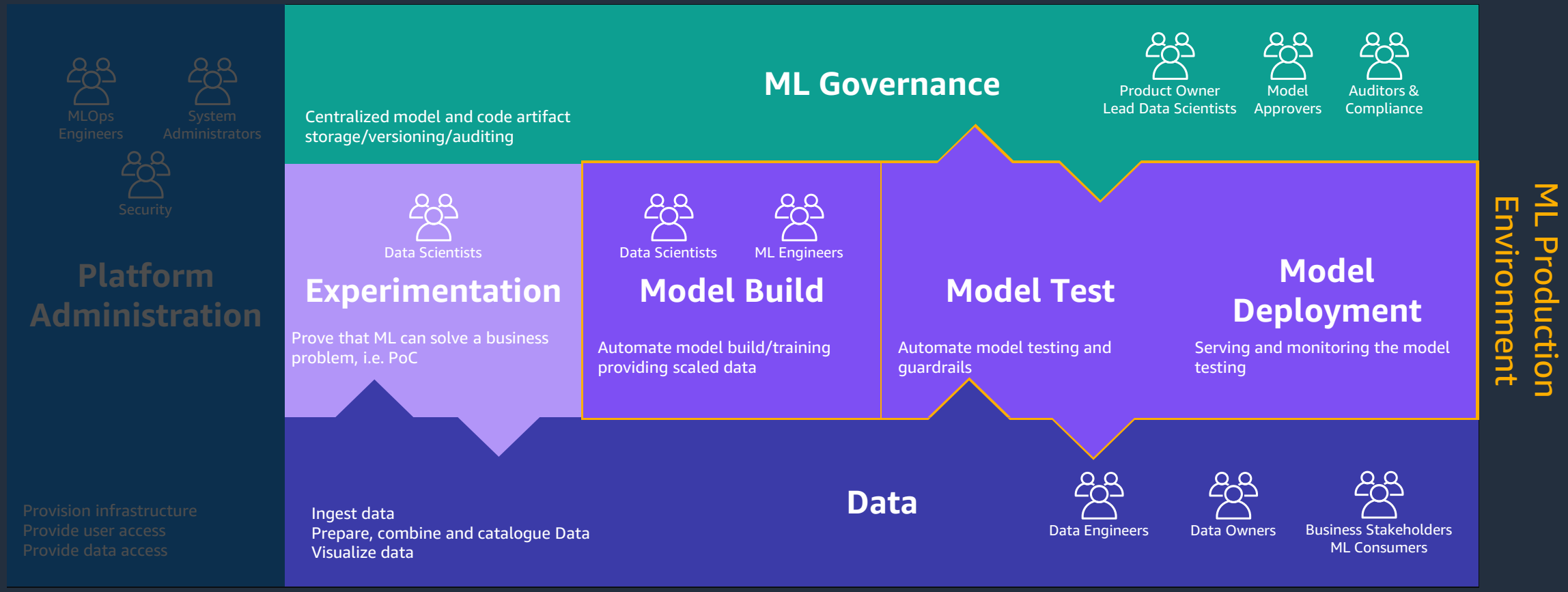
MLOps Foundation People & Processes

RELIABLE PHASE



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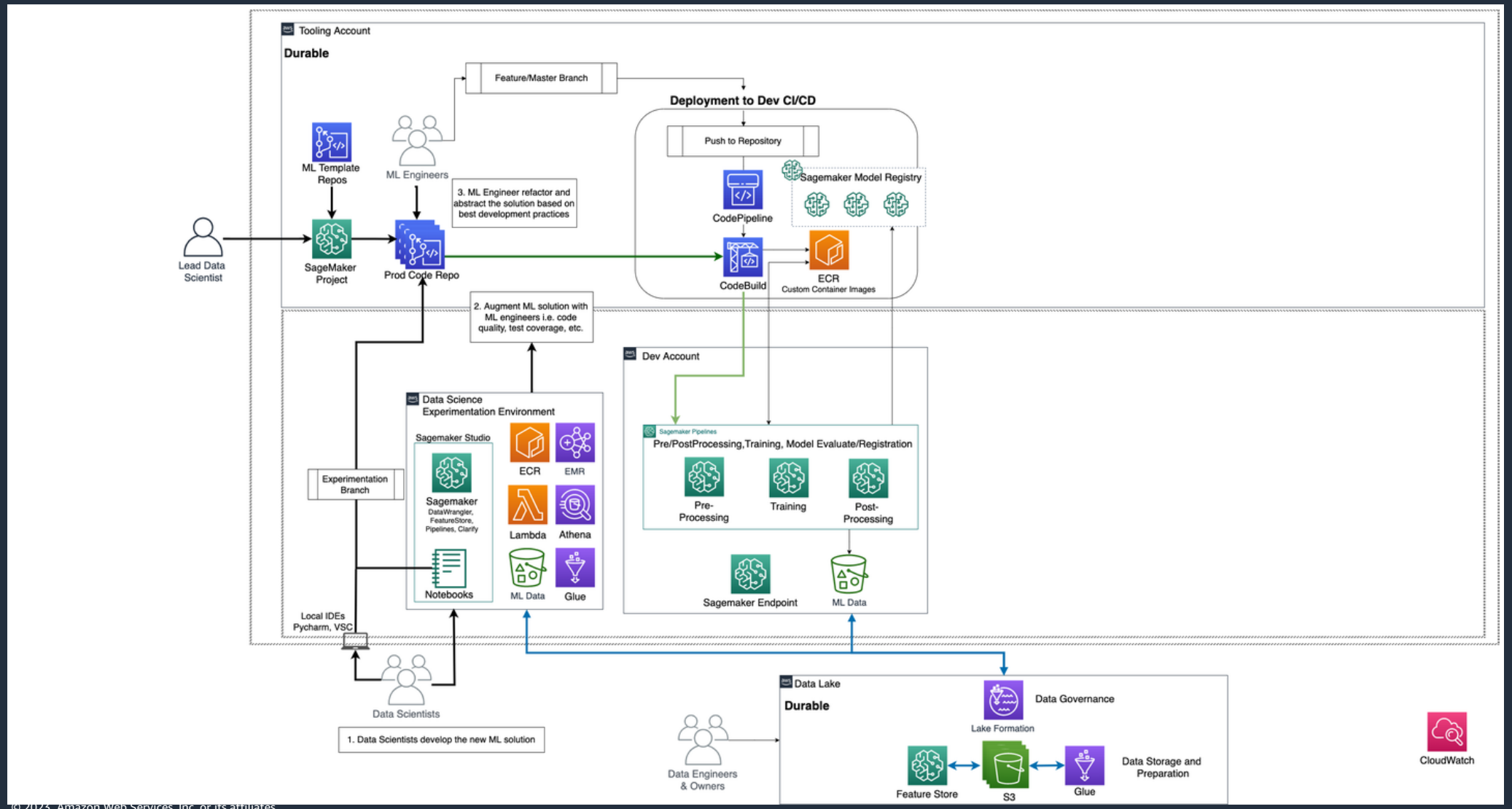
MLOPs Repeatable Phase

FROM RESEARCH NOTEBOOKS TO ML PIPELINES & AUTOMATION



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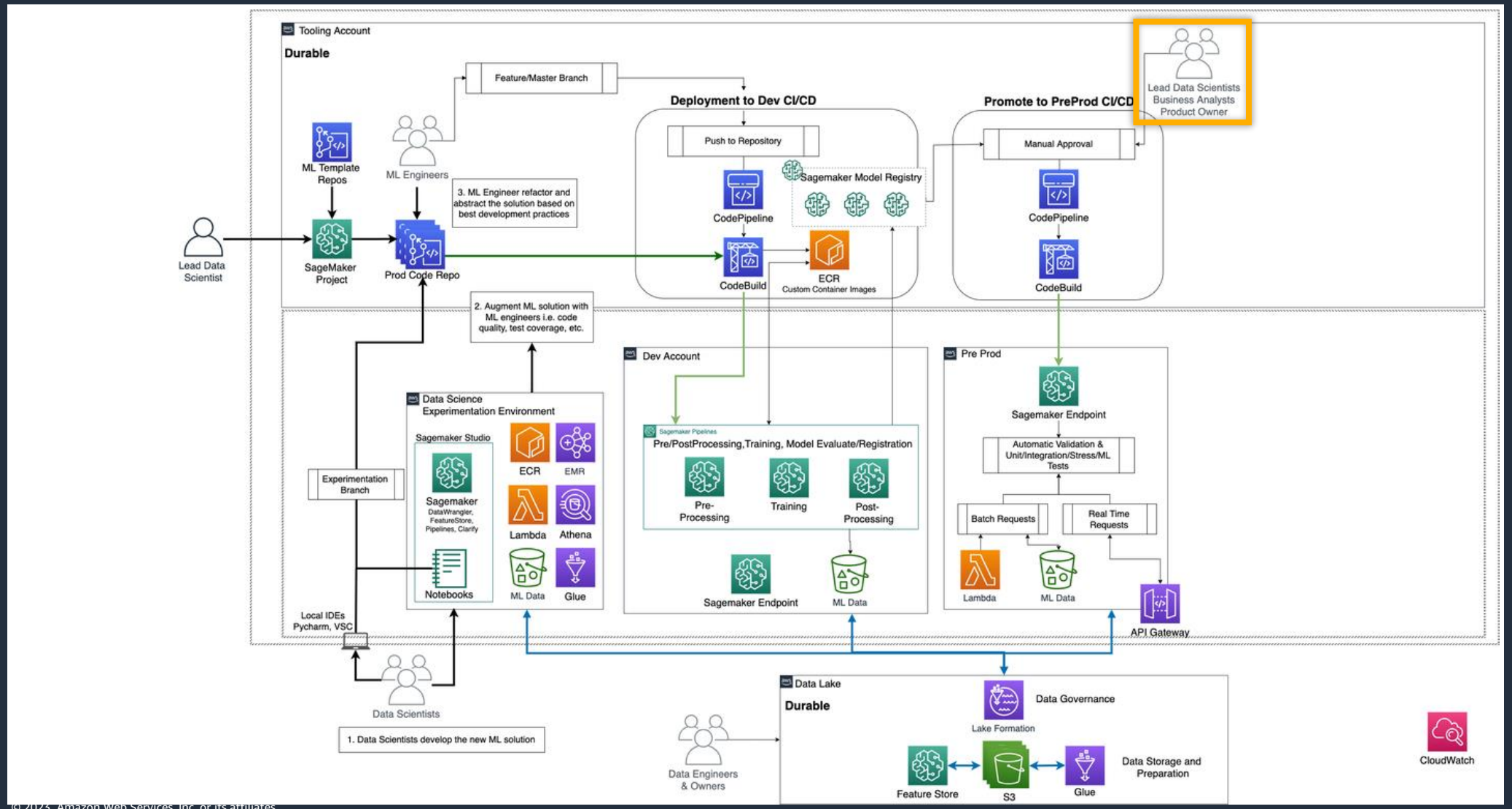
MLOPs Reliable Phase 1/2

UNDERSTAND THE NEEDS OF MLOPS & INTRODUCE AUTOMATED TESTING



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MLOPs Automatic Testing

UNDERSTAND THE NEEDS OF MLOPS & INTRODUCE AUTOMATED TESTING

Model Build and Deployment Testing

Unit test

Static code, Identify bugs, Standardize code Syntax errors

Integration test

Data, Modelling, Orchestration, 3rd Party tools integration

Stress/Load test

Latency and throughput testing of model performance

End-to-end test

Smoke test for End-to-End functionality

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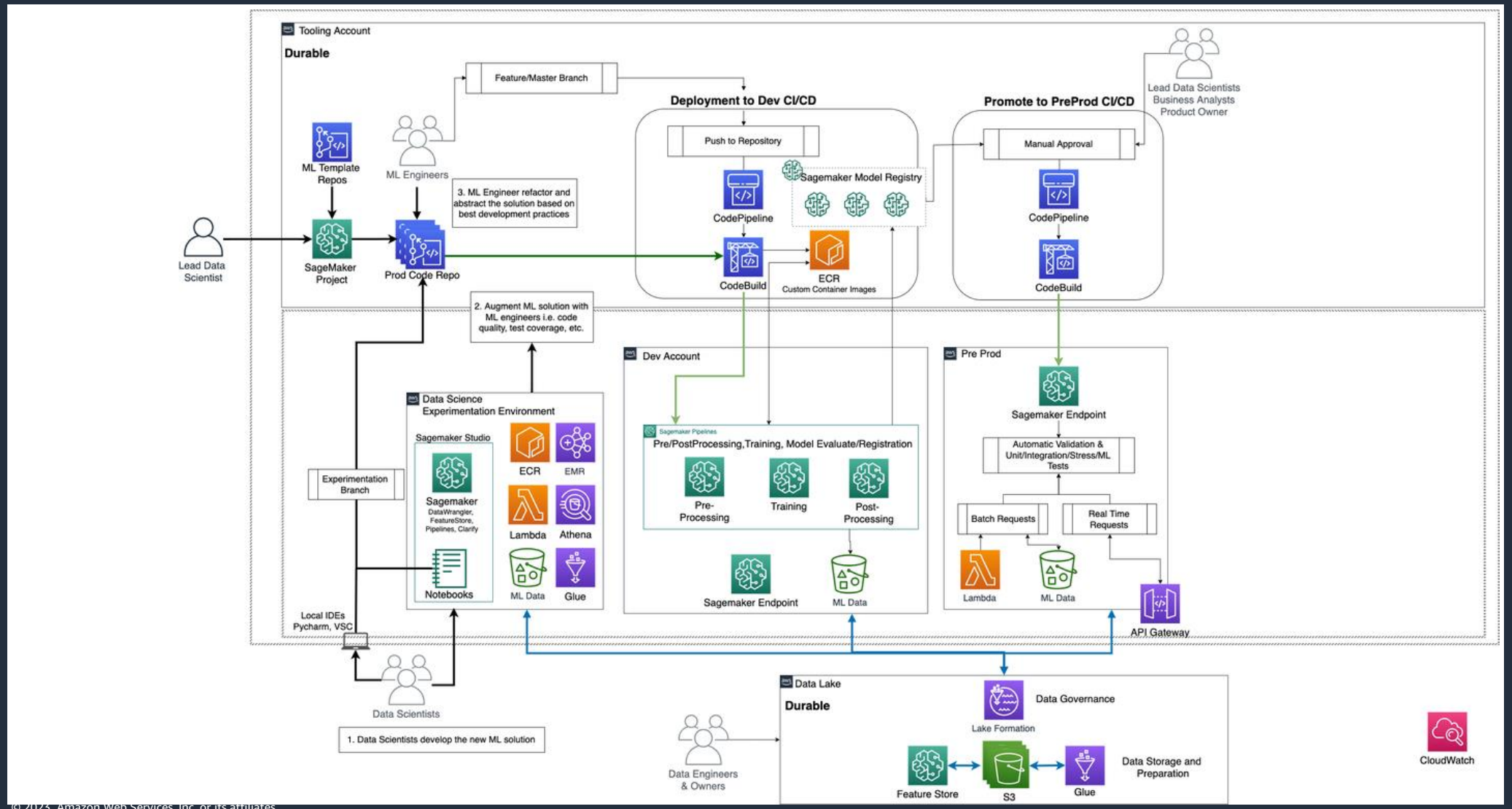
MLOPs Reliable Phase 1/2

UNDERSTAND THE NEEDS OF MLOPS & INTRODUCE AUTOMATED TESTING



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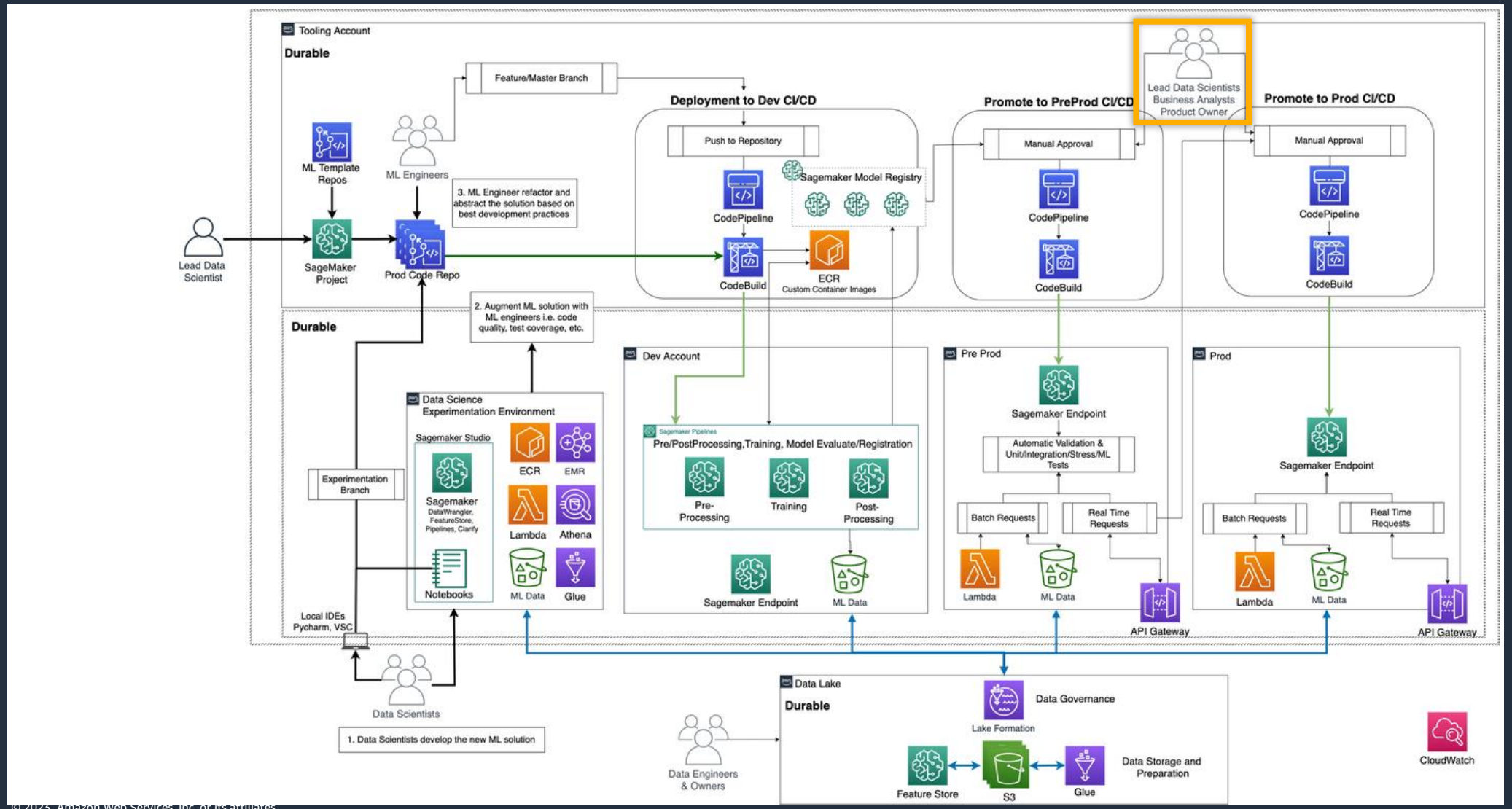
MLOPs Reliable Phase 2/2

BEST PRACTICE ON MLOPS, ROBUST AND SECURE MODELS



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MLOPs Automatic Testing

UNDERSTAND THE NEEDS OF MLOPS & INTRODUCE AUTOMATED TESTING

Model Build and Deployment Testing

Unit test

Static code, Identify bugs, Standardize code Syntax errors

Integration test

Data, Modelling, Orchestration, 3rd Party tools integration

Stress/Load test

Latency and throughput testing of model performance

End-to-end test

Smoke test for End-to-End functionality

Production Testing

Shadow test

Validate model versions in production without returning its outputs to users

A/B test

Validate model versions in production serving both outputs to users

Deployments guardrails

Securely update models in production with quick rollback functionality

Data and Model Monitoring

Continuously monitor for data and model performance drifts

aws

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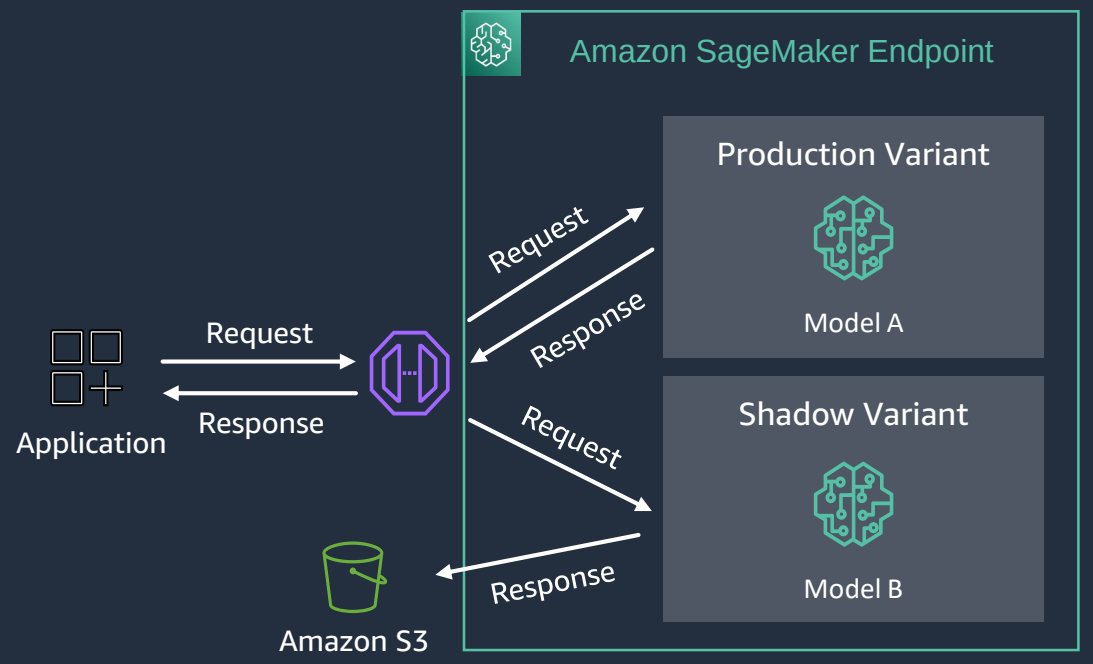
Shadow testing for SageMaker Endpoints

TESTING YOUR MODEL WITH PRODUCTION DATA AS A SHADOW ENDPOINT



- Validate performance of your models by comparing them to production models
- SageMaker takes care of mirroring requests
- Start small and dial up to control costs
- Catch potential configuration errors and performance issues before they impact end users
- Monitor progress of the shadow test and performance metrics such as latency and error rate through a live dashboard

Accessible through AWS console, CLI, APIs

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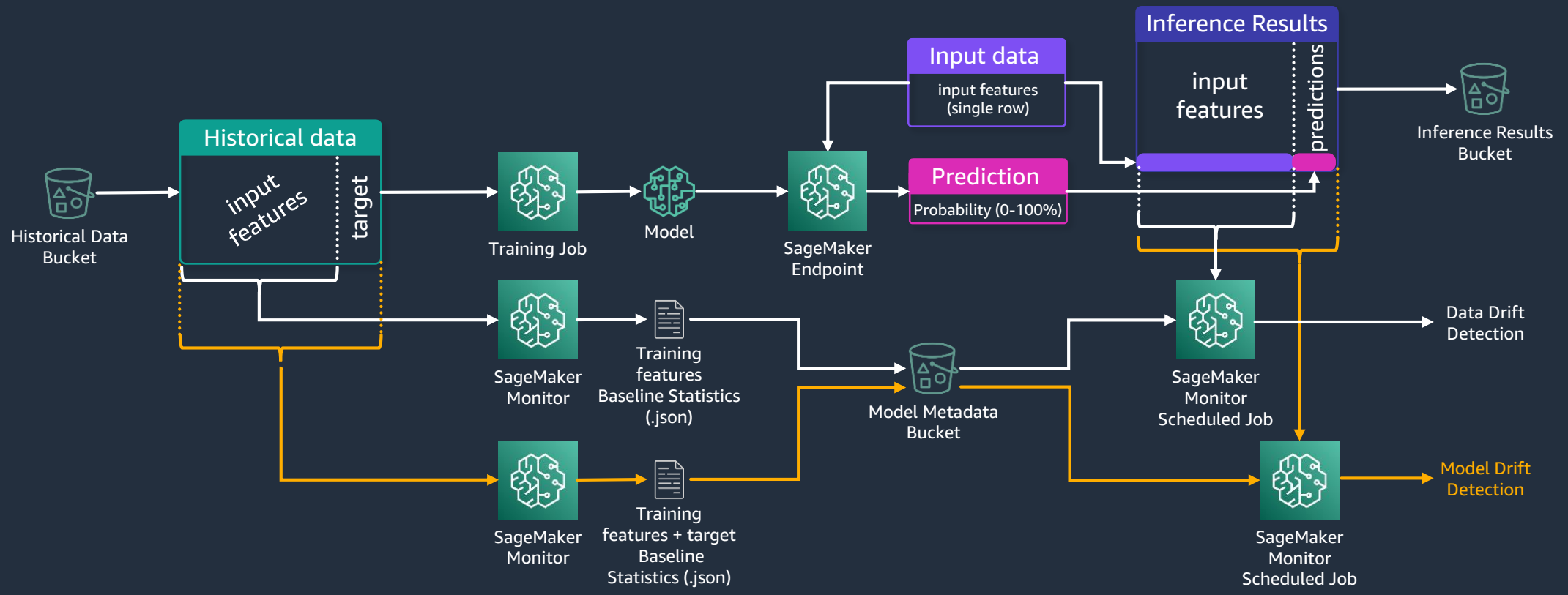
Amazon SageMaker Model Monitor

AUTOMATICALLY DETECT DATA AND MODEL QUALITY DRIFTS



https://github.com/aws/amazon-sagemaker-examples/tree/main/sagemaker_model_monitor

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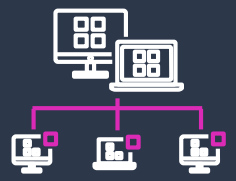
ML Governance – Model Dashboard

UNIFIED MONITORING ACROSS ALL YOUR MODELS IN PRODUCTION

Business Product Owners

Lead Data Scientist



SageMaker Model Dashboard

Gain visibility into ML models through a single pane

SageMaker Models

SageMaker Model Monitor (Real time & Batch)

SageMaker Clarify

Model name	Risk rating	Model quality	Data quality	Bias drift	Feature attribution drift	Endpoints
[Model-01]...	High	1/1/2022 5:30PM	1/1/2022 5:30PM	Inactive	1/1/2022 5:30PM	Endpoint_Name
[Model-02]...	Unknown	1/1/2022 5:30PM	1/1/2022 5:30PM	1/1/2022 3:00PM	1/1/2022 10:00AM	Endpoint_Name
[Model-03]...	High	1/1/2022 5:30PM	Scheduled	Scheduled	1/1/2022 10:00AM	Endpoint_Name & 3...
[Model-04]...	High	None	None	None	None	Endpoint_Name
[Model-05]...	High	None	None	None	None	Endpoint_Name & 3...
[Model-06]...	Medium	-	-	-	-	-
[Model-07]...	Medium	Inactive	Inactive	Inactive	Inactive	Endpoint_Name
[Model-08]...	Medium	Inactive	Inactive	Inactive	Inactive	Endpoint_Name & 3...
[Model-09]...	Medium	Inactive	Inactive	Inactive	Inactive	Endpoint_Name
[M...		cheduled	Scheduled	Scheduled	Scheduled	
[M...		cheduled	Scheduled	Scheduled	Scheduled	
[M...		cheduled	Scheduled	Scheduled	Scheduled	
[M...		cheduled	Scheduled	Scheduled	Scheduled	
[M...		cheduled	Scheduled	Scheduled	Scheduled	
[M...		cheduled	Scheduled	Scheduled	Scheduled	

Model Card (Risk Rating)

SageMaker Lineage Tracking

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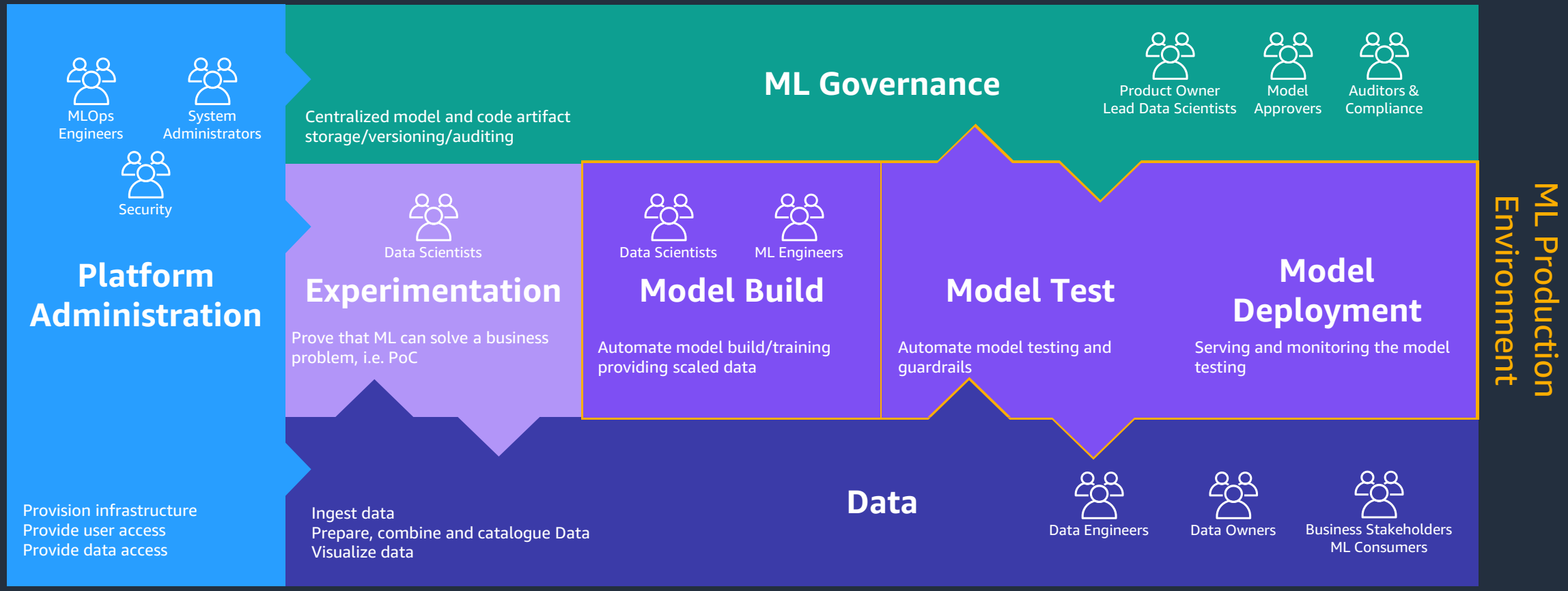
MLOps Foundation People & Processes

SCALABLE PHASE



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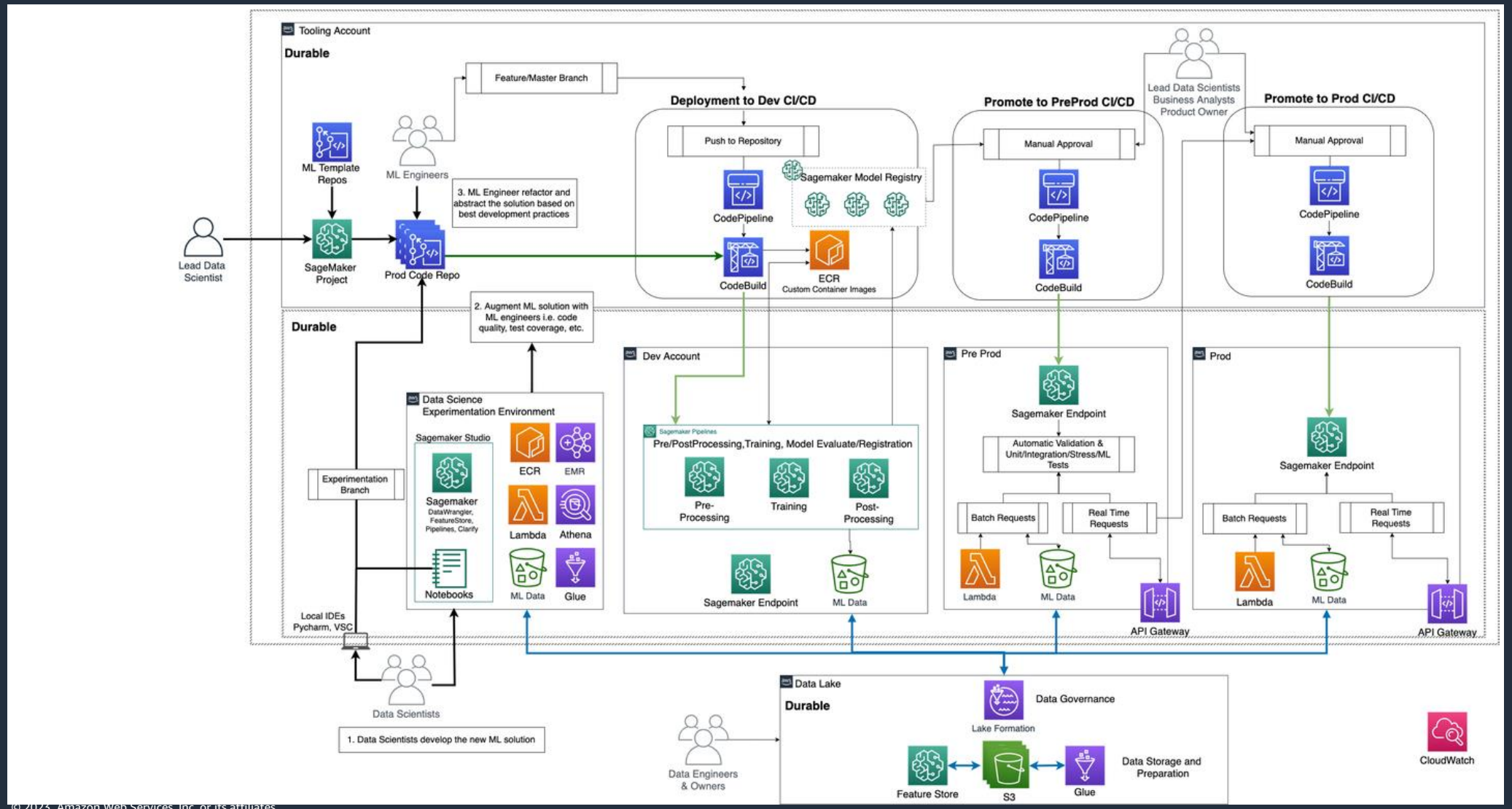
MLOPs Reliable Phase

BEST PRACTICE ON MLOPS, ROBUST AND SECURE MODELS



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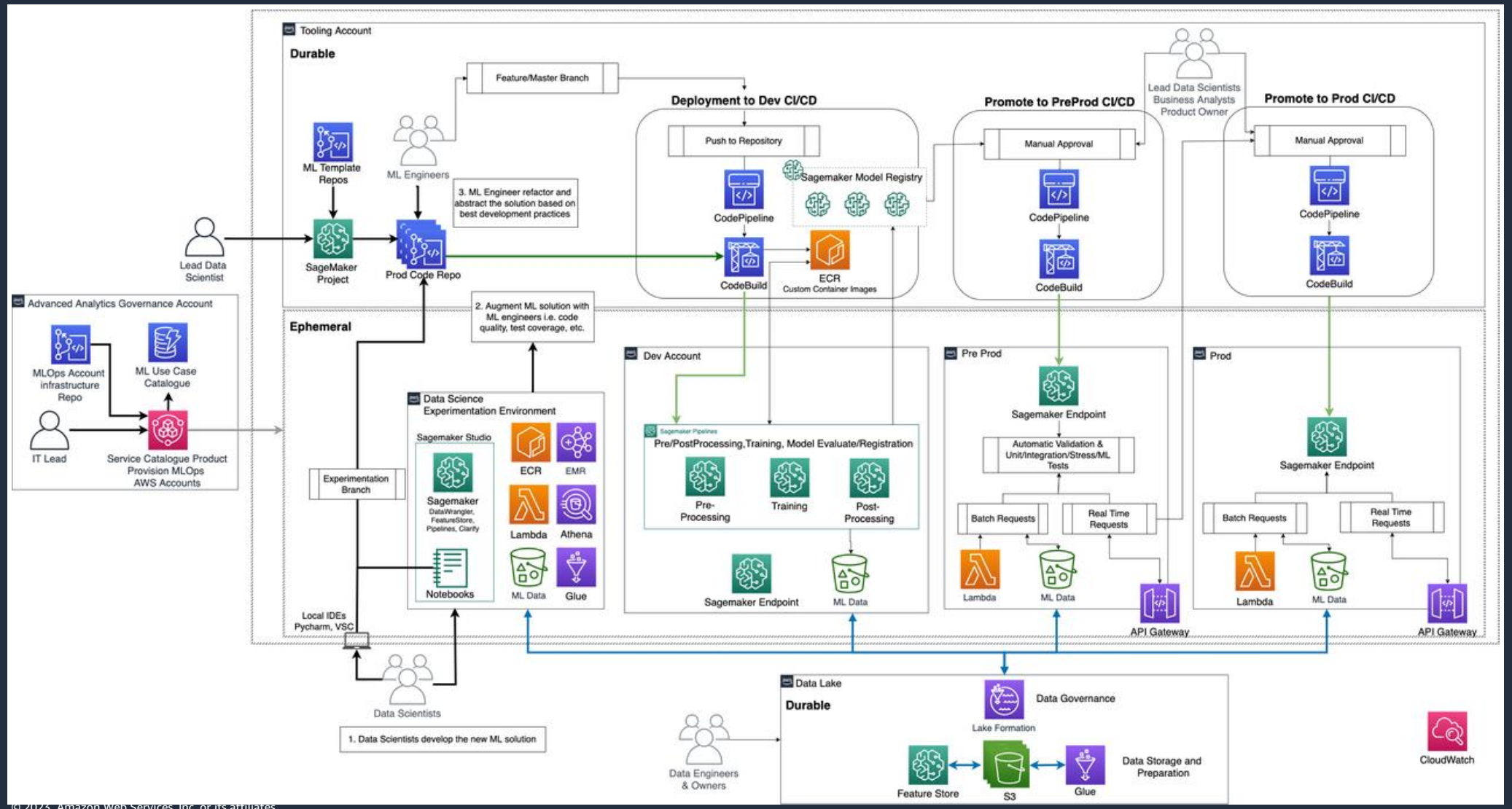
MLOPs Scalable Phase

MULTIPLE TEAMS AND ML USE CASES ADOPT MLOPS



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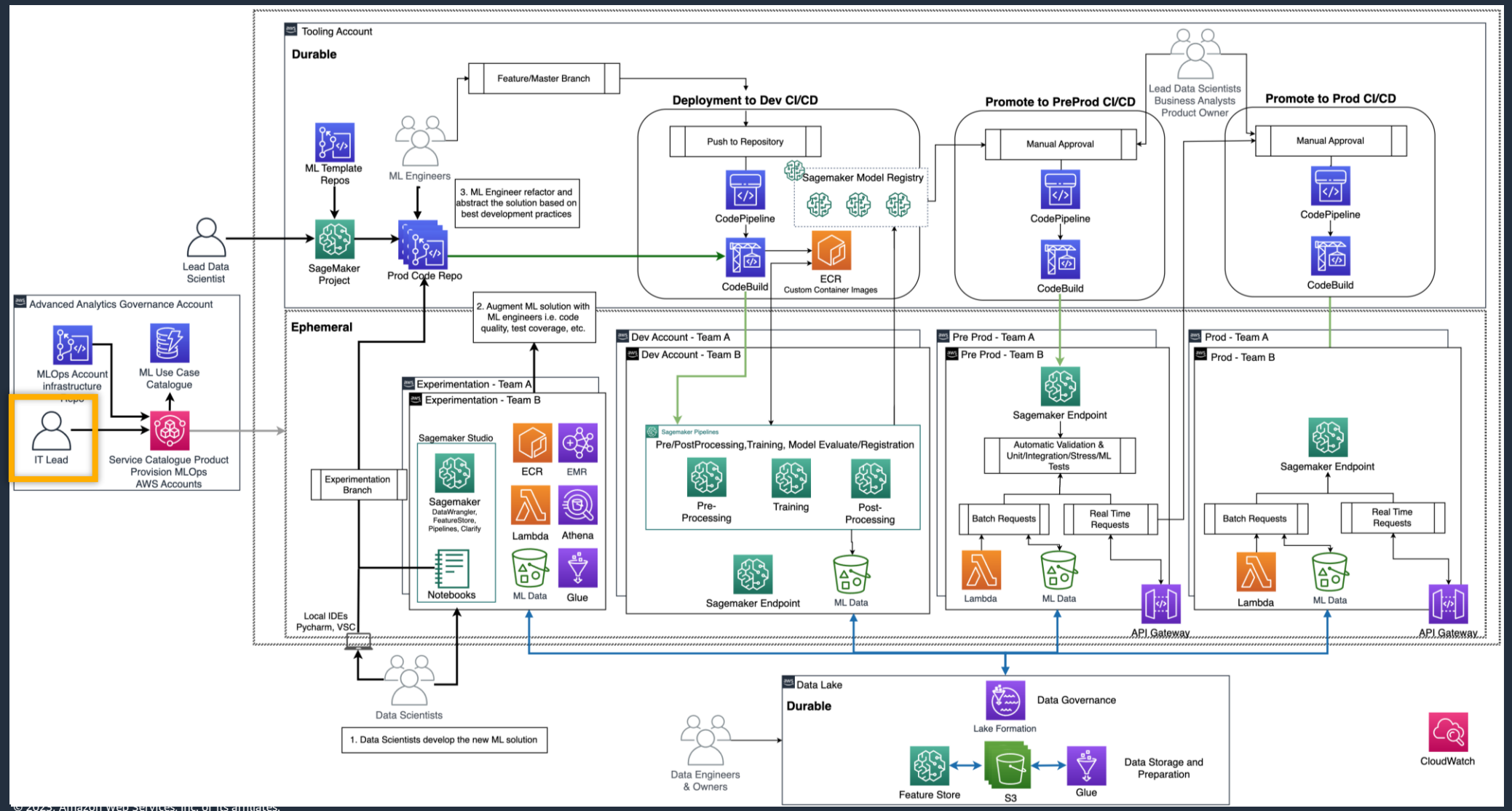
MLOPs Scalable Phase

MULTIPLE TEAMS AND ML USE CASES ADOPT MLOPS



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MLOps Capabilities, Services, & Benefits

SCALABLE PHASE

	Capability	AWS Service	Benefit
1	Experimentation	Amazon SageMaker Studio Amazon SageMaker Experiments	Standardization of data science research development and centralized experiment tracking
2	Model Build	Amazon SageMaker Processing Amazon SageMaker Pipelines Amazon SageMaker Model Registry Amazon SageMaker Project	Standardization of development, automation, model versioning control, and ML use case templates – Speed to Value
3	Model Testing & Deployment	Amazon SageMaker RT Endpoints Amazon SageMaker BatchTransform or equivalent Amazon SageMaker Shadow Testing Amazon Sagemaker Model Monitor	Scalability, low latency/high throughput for real time, and automatic deployment guardrails - Robustness
4	Model Governance	Amazon Sagemaker Model Cards & Dashboards	Centralized and efficient model auditing, monitoring, and alerting of model/data drifts, - Compliance
5	Platform	Service Catalogue Amazon SageMaker Custom Projects Templates	Reduce MLOps environment instantiation time - Minimized Time to Onboard a New Team
6	Data	AWS LakeFormation AWS Glue Amazon EMR Other Services	Automate data ingestion, transformation, and access control – Data Quality

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The Journey to Scale MLOps

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ML Use Cases at Scale – Timeline

PRIOR THE MLOPS PLATFORM DELIVERY

ASSESS & PLAN

MOBILIZE & DEPLOY

SCALE & MODERNIZE

Diagnostic Process

AWS understands Customer needs, priorities, and dependences

Training & Enablement

AWS upskills Customer in three key pillars: Technical, Business, and Delivery

Shape a Common Vision across Customer BUs

AWS co-design with Customer the roadmap to success, including timelines, deliverables, checkpoints, and KPIs

Execution Phase

AWS co-develop and co-deliver the sustainable MLOps platform and productionize the first ML use cases

AWS Service Team, AWS Account, Specialist, and Support Team, AWS Professional Services, & Partners

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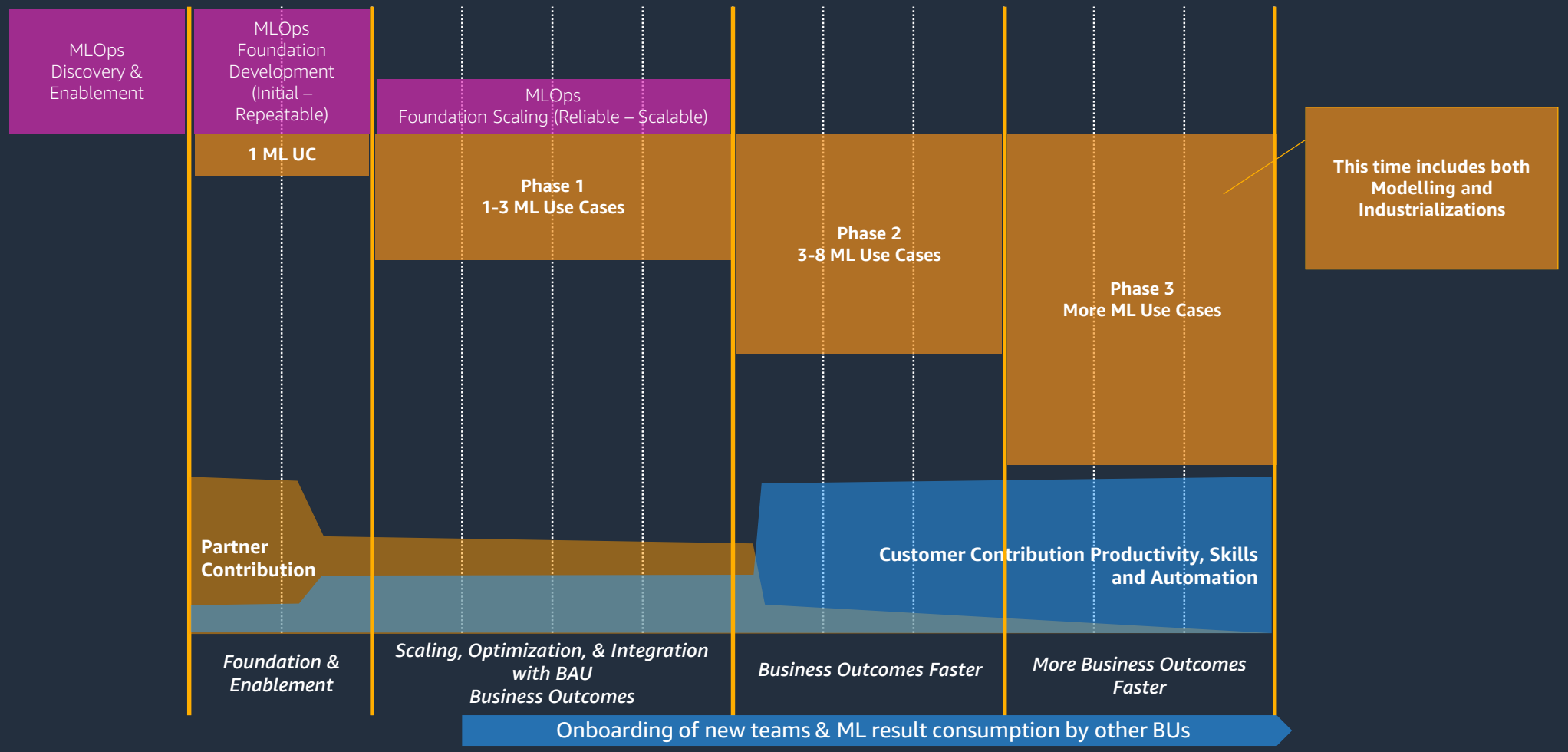
MLOps ML Use Cases at Scale – Timeline

PHASED MLOPS PLATFORM DELIVERY BASED ON PRIORITIZATION



** Time of ML use case varies depending on their complexity but industrialization time decreases exponentially e.g. from weeks to hours*

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MLOps Resources

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Public Resources

- MLOps
https://aws.amazon.com/sagemaker/mlops/
- Amazon SageMaker documentation
https://docs.aws.amazon.com/sagemaker/index.html
- MLOps Foundation Roadmap for enterprises
https://aws.amazon.com/blogs/machine-learning/mlops-foundation-roadmap-for-enterprises-with-amazon-sagemaker/
- Multi-account MLOPs Platform Source Code Examples
https://github.com/aws-samples/sagemaker-custom-project-templates/tree/main/mlops-multi-account-cdk
- How NatWest Group built a scalable, secure, and sustainable MLOps platform
https://aws.amazon.com/blogs/machine-learning/part-1-how-natwest-group-built-a-scalable-secure-and-sustainable-mlops-platform/
- BP Scales Its Data Science Machine Learning Operations on AWS
https://aws.amazon.com/solutions/case-studies/bp-machine-learning-case-study/

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Thank you!

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